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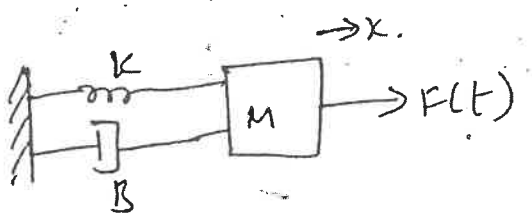
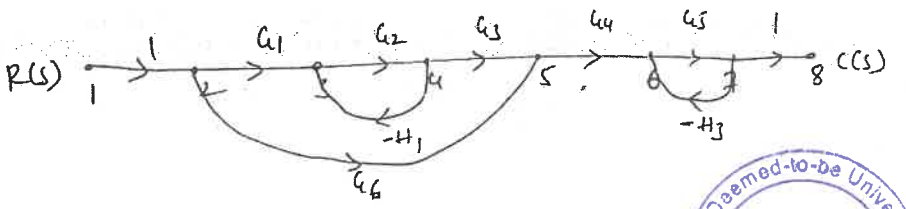
FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

CONTROL SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**05 x02=10**
1.	**a.**	In force voltage analogy, mass and spring are replaced by which electrical elements?	L2	CO1	**02**
	b.	Define the following terms: **i.** Peak overshoot **ii.** Peak time	L2	CO2	**02**
	c.	State angle and magnitude criterion of root locus.	L2	CO4	**02**
	d.	Define: **i.** Gain margin **ii.** Phase margin	L2	CO5	**02**
	e.	Describe the significance of stability in control system design.	L2	CO6	**02**
		PART-B			**06 X 05=30**
2.	**a.**	Using an appropriate block diagram, analyze the temperature control system.	L3	CO1	**05**
		OR			
	b.	Compare open loop and closed loop control systems.	L4	CO1	**05**
3.	**a.**	Obtain the differential equations for the mechanical system shown in figure 3a.  Figure 3a	L4	CO1	**05**
		OR			
	b.	Determine the overall transfer function of the system using Mason's gain formula whose signal flow graph is shown in figure 3b.  Figure 3b	L3	CO2	**05**

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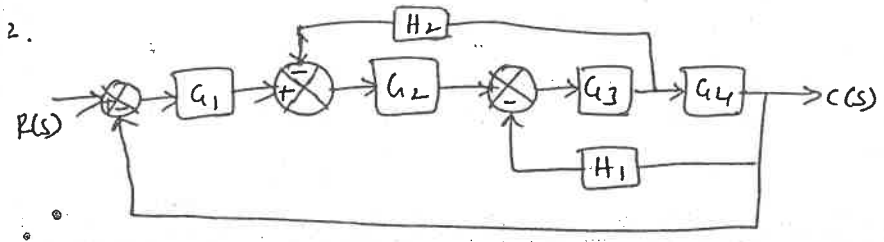
4.	a.	Derive an expression for rise time for a second-order system with unit-step input.	L3	CO3	**05**
		OR			
	b.	A unity feedback control system as the forward path transfer function $G(S)=K(2S+1)/S(4S+1)(S+1)^2$. The input $r(t) = 1+5t$ is applied to the system. It is desired that the steady state value is ≤ 0.1. Find the value of K.	L4	CO3	**05**
5.	a.	Utilizing the characteristics equation provided below, evaluate the system's stability. $$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$$	L4	CO4	**05**
		OR			
	b.	Explain the rules for the construction of the root locus plot.	L2	CO4	**05**
6.	a.	Determine the frequency domain specifications for a second order system with unity feedback and $G(S)=225/[S(S+6)]$.	L4	CO5	**05**
		OR			
	b.	Explain, how gain margin and phase margin are determined from Bode plot.	L2	CO5	**05**
7.	a.	Identify the control of the fuel-air ratio in an automotive engine contribute to engine performance and emissions control?	L3	CO6	**05**
		OR			
	b.	Classify the key components involved in the control of the read/write head assembly of a hard disk?	L4	CO6	**05**
		PART-C			**06 X 10=60**
8.	a.	Obtain the transfer function of the feedback control system shown in figure 8a by block diagram reduction technique.  Figure 8a.	L3	CO2	**10**
		OR			
	b.	Write the differential equation for the mechanical system shown in figure 8b and also draw electrical analogous circuit based on F-V and F-I analogies.	L3	CO1	**10**

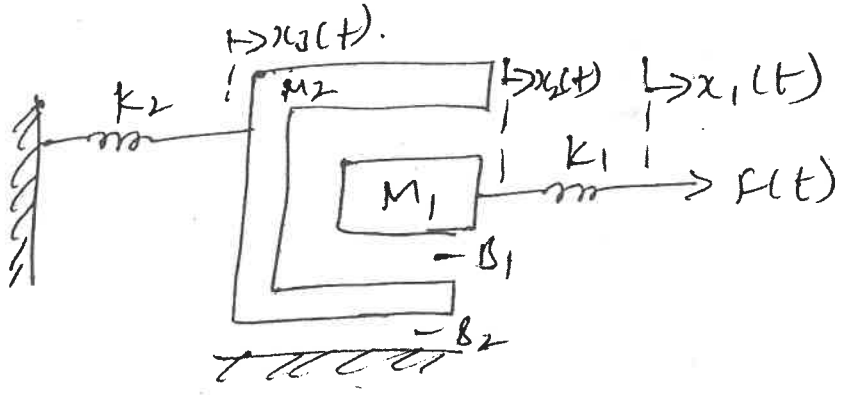


Figure 8b

9.	**a.**	Derive the expression for response time of under-damped second order system for unit step input.	L3	CO3	**10**
		OR			
	b.	A negative feedback system has the following function. G(S)=9/[S(S+2)], H(S)=1. Determine its natural frequency, damping ratio, damped frequency of oscillation, damping factor, rise time, peak time, % peak overshoot, and approximate 5% settling time. Assume unit step input.	L4	CO3	**10**
10.	**a.**	Find K_P, K_v and K_a for a system having $G(s)=\frac{(s+10)}{s(s^3+7s^2+12s)}$ Evaluate steady state error when the input r(t) is given by **i.** r(t) = 5u(t) **ii.** r(t) = 2tu(t) **iii.** r(t) = $4t^2u(t)$	L4	CO4	**10**
		OR			
	b.	For a unity feedback system $G(s) = \frac{K}{s(s+1)(s+3)}$ .Sketch the complete root locus and comment on the stability of the system.	L4	CO4	**10**
11.	**a.**	The output of a control system is related to its input by $$G(s) = \frac{K(S+10)}{S^2(S^2+2S+10)}$$ **i.** Find the value of k, so that the steady state error for a unity parabolic input is ≤ 0.1 **ii.** Verify the closed loop system is stable or not.	L4	CO4	**10**
		OR			

	b.	Sketch the Bode plot for the following transfer function and determine the system gain crossover frequency, phase crossover frequency, gain margin and phase margin. $$G(s)H(s) = \frac{80(s+5)}{s^2(s+50)}$$	L4	CO5	**10**
12.	**a.**	Construct the bode plot for the function G(S)=80/[S(S+2)(S+20)]. Determine phase margin, gain margin, phase cross over frequency, gain cross over frequency. Comment on stability of the system.	L4	CO5	**10**
		OR			
	b.	Inference a case study detailing the design and implementation of a control system for the regulation of the fuel-air ratio in an automotive engine?	L4	CO6	**10**
13.	**a.**	Compare and contrast the control challenges between automotive engine fuel-air ratio control and hard disk read/write head assembly control.	L4	CO6	**10**
		OR			
	b.	Inference a case study detailing the design and implementation of a control system for the control of the read/write head assembly of a hard disk drive?	L4	CO6	**10**